First, I would like to thank my family, you taught me a love that is whole from the moment I was born and I have lived from my heart ever since. You guys are the source of my optimism in the face of climate change, so thank you!

To Lucas, you are an inspiration in the way that I learn about climate change. I'll always remember that night in the library when you saw the very last chapter of your book about climate change called "Loving"; you drew an arrow out, writing "FEDE" in all caps. I always told you love was part of this!

And to Tommy, thank you for inspiring all of my loving. You have shown me a love that spans lifetimes.

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INTRODUCTION

This final work is dedicated to researching climate change through scientific and economic lenses, unified by an overarching theme of love. It will reflect on the relationship between humans and Earth, and how this dynamic dictates if well-being is predominantly measured by economic growth and whether such growth takes an exploitative nature. As stated by Ailton Krenak of Krenak ethnicity, the imbalance between humanity and the natural world comes from seeing human beings as the measure of all things and failing to value nature for just being.

> "Everything here is alive, the deer, the morning breeze, the grass, the mountain. You want to buy this land that is sacred to us. Do you not understand that you will die and be buried in it?"[1]

Humanity's relationship to the Earth varies greatly across the globe due to many factors such as endowment of natural resources, temperature, history, technological and industrial development, population density, and countless others. The main channel discussed in this paper, that arguably determines the nature of the relationship between humans and the Earth, is Indigeneity. Indigeneity has a diverse understanding; it also varies over time and across the globe. The understanding followed in this paper is broad but primarily agrees with the definition that a community is Indigenous if it is rooted through generations, with a continuity dating back to pre-colonial times, to the natural world that they live on. First introduced will

[1] (Guerra, 2021).

be the Kwakwaka'wakw, Indigenous to the Pacific Coast of the Northwest of the United States. Second, come the Krenak, Indigenous to Southeast Brazil. And thirdly, are the iTaukei, Indigenous to the Serua Island of Fiji. As a result of geographical location to a coast or to the Global South, these are among the communities that are most vulnerable and already experiencing the worst of the effects of climate change. Therefore, adaptation and mitigation strategies have become almost embedded in that same culture, having emerged out of absolute necessity for survival. Covered in this paper will not only be the relevant stories and customs of Indigenous communities, and the now embedded tradition of adaptation to survive, but also their solutions to effectively do their part to prevent climate change. All goes to maintain that when drafting global strategies to adapt and mitigate climate change, Indigenous science must be seriously taken into account. Thus, this paper will do its part to clear out the stigmatized reservations surrounding Indigenous knowledge, often discounted in the academic and scientific spheres. Rather, the Indigenous example of a mutually loving relationship with the Earth, as well as Indigenous science, must be brought to the frontier of environmental research and solutions.

So, how does love come into this fight? In discussion of the effects, causes, and real solutions of climate change, for as long as humans lack the ability to empathize with our humanity all the way on the other sides of the globe, globally coordinated action will not come. Within this same idea, for sustainability to be the dynamic and therefore intergenerational concept that it is, it requires that we preserve the Earth and the standards of living reaped today for future generations as they come. For this to be possible, we, the humans of today, must see past ourselves and extend

our care far enough to those we cannot even name yet. That is how love comes into this fight.

RESEARCH AND METHODOLOGY

This paper maintains a global standpoint, with specific references to the United States, Brazil, and the Republic of Fiji. The time frame is vast, pulling from centuries-old Indigenous knowledge and tradition, as well as anecdotes, theories, and reflections from the most relevant scholars and thinkers (Indigenous and not) in climate change research today. Reference and inspiration comes from countless sources, most notably the books of Ailton Krenak, Malcom Ferdinand, and the PhD dissertation of Dr. Lyla June Johnston. These works, and the rest of the consulted sources are detailed in the bibliography where they are organized by chapter.

The first chapter of this literature review is called "Indigeneity", beginning with a discussion of its definition, and then bringing nuance to how Indigeneity might shape the nature of the relationship between people and the Earth. The second chapter is called "Kincentric Ecology" which essentially discusses an understanding of love in the context of climate change. And finally, the final chapter is called "Solutions From Indigenous Science", which presents various solutions to climate change that are directed by Indigenous communities.

CHAPTER ONE – INDIGENEITY

1.1 Defining Indigeneity

What makes a person Indigenous? "Indigeneity" has many understandings, varying and transforming over time and space. There are around 370 million[2] Indigenous people in the world, but not all fall under this category for the same reasons. The term's Latin origin *indigena* means "native" or "sprung from the land"[3] and has been used since as early as the year 1588. Building on its root, *Indigenous* was used to name the people who first occupied a specific area. When colonial regimes prevailed all over the world, this recognition was crucial so as to separate the colonial powers from those forcefully colonized. Colonial thinking is dangerous because it carries with it a sense of entitlement to natural resources, peoples' rights, and land. Through tradition and varying degrees of radical isolation from "mainstream communities", Indigenous people are characterized by a strong ethnic and cultural identity that is protected for future generations to belong. This history of colonization has shaped and transformed the term Indigenous because it challenges the cultural persistence of a community and propels it into action to fight for their collective identity and remembrance.

> "There is a difference between being shy and being selective. [...] Sometimes when someone is avoiding you, they are just avoiding you. We have the right to be obscure. It is not an invitation to colonize us. It is not seduction. Boundaries can be so beautiful. Teach so much. This is a whale that is Black but sometimes wears enough algae to appear brown or orange at sea on the

[2](Oxford Bibliographies, 2018)
[3](Etymonline, 2023)

move. Once a group of these whales who were being followed by scientists stayed underwater for an hour, traveled four miles without surfacing. Do you think the researchers got the hint?" [4]

Continuous and pre-colonial occupation of a certain area of land is usually an indicator of Indigeneity, however also recognized are people and communities who may have been displaced or forced to search for other land to live, for their own safety to avoid erasure and ethnic cleansing. It does not make them non-Indigenous, but survivors. And again, Indigeneity is extremely complex. It can also be that firstness, or aboriginality, does not necessarily make a community Indigenous. Overall, it is clear that the global identity of Indigeneity is one that can be confusing. And so, the understanding followed in this paper will be the one implied from the word's Latin root: Indigeneity means to be born and tied through history, tradition, and practice to a home land.

1.2 Theories on Colonization — A Non-Indigenous Relationship to the Earth

The following section reflects on coloniality and how it acts as a channel through which Indigeneity to a land is interrupted and thus a non-Indigenous relationship to the Earth is formed.

1.2.1 Malcom Ferdinand's "Double Fracture"

[4](Gumbs, 2020)

Malcolm Ferdinand, author, lecturer, and environmental (water) engineer, is one of the most prominent activists in intersectional environmental justice and is thus a great place to start to develop the disconnection between colonialism and the natural environment. In terms of geographical scope, Ferdinand's work is concentrated in the Caribbean and more specifically his home island of Martinique, from which he derives many of his primary mentors in thought and his strongest motivations for his work in decolonization that he continues today. While this scope does not match that of this literature review, his work in condemning the effects that colonization has had on the environment is extremely relevant to understand how deep the roots of this crisis go.

In his book, *Decolonial Ecology: Thinking From The Caribbean World* (Ferdinand, 2019), he presents this concept of the double fracture: colonial and environmental, providing a theoretical constant through which we can perceive and visualize the (dis)connection between the two.

1.2.2 An Earth Enslaved – The Environmental Fracture

It is undeniable that for decades now rapid digitalization, urbanization, and the economic goal of an ever-growing economy have led to the depletion of natural resources and destruction of Earth, so viciously that nature's regenerative abilities are also degrading. The explanation of how we got here begins with an underlying value system that places human goals over the well-being of the natural environment. The environmental fracture, as defined by Malcom Ferdinand, describes the separation between the Earth and humankind, the natural versus the

man-made. It is this first distinction that laid the seeds for the climate crisis that now threatens the survival of all of humanity. The destructive consequences from decades of exploiting the natural Earth and its offerings are cumulative and will not reduce without serious intervention and a radical transformation in the principles on which the global economy operates.

However, Ferdinand argues that the ecological critique alone fails to reflect on the deeper roots of climate change – racial discrimination and colonization. In an era of mass-consumption and global markets, most populations have embraced the industrial experience of growing up in urban cities that come with a heavy separation between people and nature. This reduction of nature to resources to be extracted, facilitated the colonial regimes that have followed. To complete the explanation of how humanity has gotten to this point, Ferdinand's second argument, the colonial fracture, is introduced.

1.2.3 The Colonial Fracture

> "Like the day the spaniards climbed down from their horses
> and asked us if they could buy the mountains." [5]

The other side of Ferdinand's double fracture is the colonial fracture, separating the colonizers, from those colonized. Colonialism was a result of greed for money, riches, and power, and thus subjected humans to a hierarchical system. In its execution,

[5](June Johnston, 2013)

colonization involves the displacement of people from their homelands and a violent appropriation of land and all of its natural resources. It is this parasitic system that has led to the existence of colonies all over the world, and has lasting consequences on the current world system, the one that faces the threat of the changing climate. Racial and environmental injustice both have roots and histories in colonialism and yet their intersection is almost never spoken of. For consistency and proper contextualization of his work, Ferdinand delivers his ideas through an analogy, demanding that we perceive modernity from the "hold of a slave ship", and that it is only from this perspective that "we cannot sustain a separation between ecological and social injustice"[6]. It is exactly this, this separation between the two experiences, that he calls the "double fracture".

Ferdinand's book is a "sutural writing"[7], invoked to highlight this intersectionality and heal this fracture between the ecological perspective and the racial perspective. By integrating the two, we develop the decolonial ecological critique which advocates for environmental justice conditional on human justice, and vice versa. One can think of intersectional environmentalism as an alliance, conceived by a colonial past that not only exploited people, but also the Earth.

Using the example of the rising sea-levels, Ferdinand represents the double fracture by explaining that the root of this danger is not the rising sea-level itself, but the layers of historical, racial, and political injustices that have charged the changing climate. Escaping this "double-fracture" means not only being proactive about

[6](Ferdinand, 2022a)
[7](Ferdinand, 2022b)

climate disasters themselves, but committing to saving all lives. Furthermore, in one of his lectures critiquing technology, Ferdinand points out that many of the most well-known French ecologists would theorize about nuclear bombs, seeming to dismiss that the one that devastated Hiroshima was constructed with uranium that came from Belgian Congo[8]. Environmental injustice and racial injustice are inextricably linked, and it is not sufficient to comment only on one side of the fracture. Ultimately, this double fracture begs the argument that the climate crisis that we find ourselves fighting today is the result, or culmination of centuries of creating a world value system that prioritizes monetary wealth and the whiteness of your skin above living, understanding, and loving.

1.3 From the Roots of Colonialism

The colonial fracture that has facilitated the domination of Indigenous and other marginalized people is the contingency on which destruction of the Earth relies. Thus, it should be clear that stopping the ecological crisis that is climate change requires a radical revolution in the way that the world currently functions. Despite the ultra-commodified, capitalistic, way of life that narratives of modernity and progress have pushed, there are other ways that we can inhabit this Earth.

Dr. Lyla June Johnston, an Indigenous musician and scholar of Diné (Navajo), Tsétsêhéstâhese (Cheyenne) and European lineages, offers a radical rethinking of the role of humanity in life on Earth. It can be easy to believe that without humans the Earth would simply be better off, but she rejects this sentiment, laughingly

[8](Ferdinand, 2020)

taking it personal, and argues that it is not humanity but the systems of exploitation and colonization that are in power that are degrading the natural Earth. Should we believe otherwise, deeming humans to be inherently destructive, we ignore the standards that Indigenous people all over the world have embodied for millenia. Dr. Johnston reflects on the continued legacy of her people, the Diné, characterizing them as a "keystone species", a species on which entire ecosystems depend. She repurposes the word anthropogenic, a word often used to attribute man-made climate change, instead to attribute the ways in which her ancestors *anthropogenically* expanded wildlife habitats and nurtured the soil under their feet for perpetuity. Nature can flourish even without human intervention, but she makes clear that her ancestors, and many other Indigenous communities, actually play an active part in guiding the natural elements of the Earth.

Contrast this with the current land-management system of wildlife preservation: the isolation of national parks. National parks and other sorts of conservation areas are a direct result of a colonial-derived, superficial relationship to the Earth where humans and nature are separate, and if nature is to be preserved it must remain untouched. Renaming these parks and turning them into a country's prized possessions, commodifies nature as a good that can be bought and put on display. In Dr. Johnston's PhD dissertation called *Architects Of Abundance: Indigenous Regenerative Food and Land Management Systems and the Excavation Of Hidden History* (2022), she describes the "living heirlooms"[9] of natural habitat, thousands of years in the making. Throughout her dissertation she steadily condemns their

[9] (June Johnston, 2022)

scientific mislabeling as "terra nullius", "virgin land", and "wilderness". One of the prevailing colonial narratives is called the "agricultural improvement theory" or the Loch perspective, which argued that ownership of land depended on if you had *worked* on it, which by European colonial standards needed to be visible through agriculture or their standards of development. Since Indigenous stewardship took a form unrecognizable to the colonizers eyes, this argument was repeatedly used to delegitimize the sustainable land-management practices of Indigenous people, calling them "primitive", and to justify the taking of land from Indigenous people for commercial use. As a result, Indigenous communities were removed from their role as stewards of the land and control over natural resources was transferred to the colonial regime. The spreading of colonial narratives and their general acceptance worked to erase the collective memory of Indigenous peoples' harmonious relationship to the Earth and establish a deep colonial fracture in the hierarchy of humankind.

CHAPTER TWO – KINCENTRIC ECOLOGY

The second chapter will begin by interpreting the concept of kincentric ecology, a term for which there are many translations. Then, it will introduce three Indigenous communities from around the globe and bring into discussion their example of living in harmony with the Earth. The goal here is to expose the real effects of Ferdinand's double fracture by exposing how the fight against environmental destruction and colonialism go hand in hand.

2.1 Love in the Context of Climate Change – An Indigenous Relationship to the Earth

> "I learned that those mountains have a name – Takukrak – and a personality."[10]

Love in the context of climate change can best be understood from an Indigenous perspective, through the shared values that they live by. While Indigenous communities have various cultural and historical differences, they are united in their reciprocal and symbiotic relationship with Mother Earth. In this relationship, there is no separation between humans and the Earth, so their idea of family is extended to flora, fauna, and all other living beings. It is with this sentiment that community is embodied and their love is extended beyond their own people.

Kincentric ecology, or kincentricity, delineates this relationship of absolute interconnectedness with all life forms, not just the human. However, the English language and Indigenous languages do not have the same purposes, and thus have different linguistic tools available. In fact, kincentric ecology is only our understanding of such a relationship but holds little significance for someone of Indigenous heritage and experience. Kincentric ecology is like the Rarámuri concept of *iwígara*, it is like the Māori concept of *kaitiakitanga*, like Diné concept of *hózhó*; it exists in some form for all Indigenous people as it is the very core of their way of life. Hózhó (*ho-ZHO*), for example, is roughly translated from Diné Bizaad to "joy of

[10](Krenak, 2020)

being a part of the beauty of all creation"[11]. It reflects a broader ethos of a people centered in connectedness, harmony, and balance with the natural Earth. In a world increasingly individualistic, where the separation between humans and Earth has been purposeful, most languages like English do not carry the words to describe such a relationship. To live in harmony with the Earth can only be learned through experience, traditions and stories. It is much more than a concept to be debated; kincentricity is an active form of loving.

Dr. Lyla June Johnston proposes four pillars underlining a kincentric relationship with the Earth:

1. Work with nature
2. Expand habitat
3. De-center humans
4. Design for perpetuity [12]

The following Indigenous communities will embody these pillars in their own spot on the Earth through various land, life, and sea management practices.

2.2 Introducing the Kwakwaka'wakw – Indigenous to the Pacific Northwest

> "What if our systems were designed to last forever?" [13]

[11](June Johnston, 2022)

[12] (June Johnston, 2022)
[13] Ibid

One of the systems fundamental for a society to survive no matter the time period is the food system. To demonstrate the concept of sustainable consumption, take a lake containing a population of fish with a 15% regeneration rate. Sustainable consumption would look like eating only a fraction of the fish population in the lake; this would allow the human population to be fed with a local and healthy food source, and the fish population to grow steadily, creating a virtually infinite food supply. Instead, when the fish population is over-fished, its population shrinks disproportionally and its rate of regeneration diminishes as well, leading to an unsustainable food system that cannot last forever. Today, our food systems align with this model of over-consumption and little consideration for future consequences on food security.

The Kwakwaka'wakw are one of the First Nations that inhabit the Pacific Coast of Northwest United States, around present-day Vancouver Island, the adjacent mainlands, and its other surrounding islands. Due to the high biodiversity in these areas, this fortune has attracted the likes of many large fishing companies from around the world who seek profit and do not wish to abide by the environmental standards that the Kwakwaka'wakw have set for centuries. A study found that around the same time that European colonial powers reached the area, clam walls dropped in number, reflecting a real historical application of Ferdinand’s double fracture where colonization displaced Indigenous people from their lands and with it, their sustainable practices. Anthropocentric management of fishing and harvesting greatly threatens the food security of Indigenous tribes like the Kwakwaka'wakw,

and risks polluting the ocean and its inhabitants (human, plant, and animal) along the coast. These exploitative corporate schemes operate from a standpoint of human sovereignty over all other life forms, and thus follow a system of profit-maximization rather than one of restraint and reciprocity. This system is inherently unsustainable since the resources on which it relies are limited and becoming more scarce.

The following example of an anthropogenically cultivated clam garden is presented in hopes to inspire a transformed food system that rests on values of sustainability and reverence for the Earth's offerings.

2.2.1 The Ancient Clam Gardens of British Columbia

The augmented clam beds, or "clam gardens" along the Northern Pacific Coast are not only beautiful, they are intentional and have been traditionally cultivated by Indigenous communities that have inhabited the area for centuries. Clams grow on the beach surface, at different intertidal heights depending on the species but usually in the low intertidal zone within 0.8 to 1.5 meters of tide. When the tides are low, much of these growth zones are exposed, and when tides are very high, all of this area goes underwater and is not suitable for clamming. The tides can vary quite a lot, leading to mildly unpredictable harvests, depending on the patterns of the ocean. So, in order to improve both abundance and predictability of clam harvests (a mutually beneficial outcome), Indigenous communities like the Kwakwaka'wakw developed a clam farming system. By building a rock wall that matches the heights of the common growth zones of all diverse clam species in the area, sediment is

caught behind with the rolling tides and a greater surface area at these crucial levels is formed, leaving more room for clam habitat.

In addition to this, the Kwakwaka'wakw practiced mindful harvesting and restraint:

> "Clams were not only sought out opportunistically, but such methods as stone removal, terracing, selective harvesting, and the regular aeration of the soil matrix allowed the Kwakwaka'wakw and other aboriginal harvesters of the region to enhance the output of clam beds. Such actions placed larger and more predictable concentrations of clams in accessible locations– proximate to large concentrations of human settlement but also within the defined and defensible territories of harvesters' clans and villages." [14]

While "man-made" and "anthropocentric" tend to have negative connotations in the sphere of climate change discussion, Indigenous anthropocentrism represents an opposite experience whereby Indigenous communities around the world nurture their surrounding land. By tending to the well-being of clam beds, the Kwakwaka'wakw are caring for the life that feeds them, and in turn are offered a more secure source of food. This tradition and work exemplifies kincentric ecology because it relies on the fundamental principles of humans as stewards of ecosystems, demonstrating a deep understanding of the causes and effects of the environment.

[14](Deur, 2015)

2.3 The Krenak – Indigenous to Southeast Brazil

The second Indigenous community discussed are the Krenak of the southeast of Brazil. Our main reference is Ailton Krenak, a Brazilian journalist, writer, philosopher, environmentalist, and leader of the Krenak Indigenous movement, who represents the Krenak struggle in the mainstream. Forcibly separated from his people at nine years old, Krenak exposes the rest of the world to the story of the Krenak people and vocally blames colonization for fueling climate change and the erasure of his family.

> "[...] it is a sensitive intelligence to understand that there are things that need to be stabilized in the Earth's organism. Colonialism does not recognize natural borders, it invades everything furiously."[15]

In the 19th century the Botocudo tribes were many, around a combined 14,000 people, but today the only tribe of their kind in Eastern Brazil are the Krenak. They originally inhabited the Atlantic Rainforest in the State of Bahia but after numerous forced displacements, they finally resettled on a 4,000 hectare reservation on the east bank of the Doce River in the State of Minas Gerais. However after decades of persecution and forced migration from their homeland, they are now drastically lower in number and are still facing a fight against erasure, with only around 200 survivors. The long struggle of the Krenak, and Indigenous communities alike, has been a fight to reclaim their land, identity, traditions and culture. It is a fight of boundaries, both social and physical, to have sovereignty over the land that they

[15] (Guerra, 2021)

cherish and take care of. It is also a fight for political representation, for human and environmental rights, in the global arena. For this Krenak recognizes and fosters alliances between Indigenous people and those in the "dominant world". This fight for recognition carries directly into the discussion of climate change.

Ailton Krenak explains a certain connectedness in the Krenak relationship to the Earth, where humans are not separate but instead part of the natural world. He emphasizes that in apparent contrast is the capitalist regime under which the world is currently captive, one that completely disconnects people from nature. The industrialization process that began during the 18th century has radically urbanized cities, creating a system of concrete jungles, built so far up we cannot even feel the ground anymore.

> "This humanity refuses to recognize that the river, now in a coma, is also our grandfather; that the mountains mined in Africa or South America and transformed into merchandise elsewhere are also the grandfather, grandmother, mother, brother of some other constellation of human beings that want to go on sharing the communal home we call Earth." [16]

In Ailton Krenak's book, *Ideas to Postpone the End of the World* (2020), he reflects on humanity's relationship to the Earth, expressing how deeply he and his people feel empathy for all living things through his observations of the rest of the world. He calls the sky his father, the river his grandfather, the mountains his mothers and

[16] (Krenak, 2020)

sisters; to the Krenak, all life is recognized as part of the same circle. This is a connection that the rest of the world has not truly experienced and by the name of his book, will likely never understand. Today, this deep connection comes at a great cost, for living in unison with the natural world means a dependency where their well-beings are intertwined.

2.3.1 The Collapse of the Mariana Dam in Minas Gerais

The Doce River, directly translated to "sweet river", runs through Brazil and holds great economical, spiritual, and cultural significance for the Krenak people. To them, the river is a living relative that is sacred and must be protected. They call him "Watu", their grandfather, with whom they share a common lifeline. The river basin is utilized for agricultural, industrial, and mining purposes, contributing to a considerable share of the country's GDP but also under the control of various companies. Samarco, a joint venture of two other companies, established an iron ore mining spot in the Doce River, constructing one of several problematic earthworks to store the extraction waste. Six months before the collapse, a worst-case scenario examination revealed a potential "liquefaction break"[17] that could kill the local residents nearby and irreversibly contaminate the river with heavy metals. Still, no preventative action was taken by Samarco. The dam collapsed in November of 2015, sending toxic orange mud flooding down the river with no warning to those living alongside. 19 people died that day[18]. The river that the Krenak once fished out of, worshiped, and swam in, will never be enjoyed the same. Ailton Krenak blames colonialism and capitalism for this "divorce"[19] between humankind and nature. It is

[17](Phillips, 2018)
[18]Ibid
[19](Krenak, 2020)

the deep environmental fracture that allowed Samarco to turn the Doce River into a commodity to be profited off of without remorse. The collapse of the Mariana dam shows a direct violation of sustainability. Sustainability would have required the Doce River to be enjoyed by all future Krenak generations to come. Is it clear now that colonial (non-Indigenous) ventures are incompatible with the environment? "The river, now in a coma,..."[20] is the Doce River.

2.4 The iTaukei of Serua Island– Indigenous to Fiji

The third Indigenous community to be introduced are the iTaukei, located in Fiji, an archipelago of more than 330 islands in the South Pacific Ocean. Its population is made up of Indigenous Fijians, called iTaukei, and Fijians of different ethnic descents. "Se" which means flower, and "Rua" which means two, Serua Island has its village located between two hills and which has a population of around 100 people. As for the local economy, fishing is the primary source of income, making the island extremely dependent on marine life and health of the ocean. Over the past two decades, the island has experienced dangerous flooding and coastal erosion, and has been recommended by their government to relocate to safer land. The iTaukei peoples' refusal to do so demonstrates the strength of kincentric ecology as the Indigenous way of life.

2.4.1 Resistance to Relocation

The iTaukei help introduce a response to climate change called adaptation. In the discussion of climate change we often discuss strategies of mitigation and

[20](Krenak, 2020)

adaptation, defined respectively as: solutions to stop climate change, and ways in which we can adapt our lives to survive in a world where the climate is warming. The type of adaptation strategies depends on the level of climate-related risk; these can look like switching to drought-resistant crops, building defenses to protect against flooding, perfecting climate-disaster warning systems, and many more. However, in areas where climate-disaster risk is exceptionally high and unpredictable, the suggested form of adaptation is actually migration. It forces people to prioritize their own lives and leave their homes and livelihoods to flee to a more secure area.

The research on the topic of climate-driven migration and involuntary mobility is growing in response to worsening climate change effects, but the iTaukei present a unique case of "voluntary immobility"[21]; despite being officially deemed at risk due to the rising sea-level, the people of Serua Island refuse to abandon their home. Understanding this choice requires understanding the complex relationship that the iTaukei have to the Earth under their feet, which this paper motivates is greatly related to being Indigenous.

Kincentric ecology in the iTaukei's language is called "Vanua", literally translated to "land", but encapsulating something much deeper. "Na qau vanua" directly translates to "not my land" but is interpreted by the iTaukei as "the land to which I belong, of which I am an integral part: the land that is part of me and sustains me"[22]. The Indigenous Fijian relationship to the Earth is a fierce one; likening their land without

[21](Yee et al, 2022).
[22](Roth, 1973)

its people to a body without a soul. Not only do the iTaukei have a deep connection to their homeland, they also have strong networks of ties and kinship between their community members. To many, it could never be an option to leave this behind. Rather than migrating and risking the weakening of their cultural ties, the iTaukei adapt to the challenges that climate change presents. Fijian adaptation strategies become embedded into their traditions, for future generations to arm themselves. Vanua is the force that grounds the iTaukei people to Serua Island.

CHAPTER THREE – SOLUTIONS FROM INDIGENOUS SCIENCE

This final chapter will be focused on discussing where to go from here.

3.1 Willingness To Act (WTA)

Economics can be broadly defined as, "the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses"[23]. It is built upon a series of assumptions, none of which take into account the fundamental flaw of why the resources in question, the inputs, are finite. In this framework, the goal of every country is to be constantly growing, developing, and increasing productive efficiency in order to remain competitive in the global economy. In pursuit of this goal, the world has been operating in an extraction-based economy that exemplifies the contrary to a kincentric economy: an anthropocentric economy. Valuing human life above all else has come to mean surpassing the natural capacities of the planetary systems in order to keep producing and consuming, instead of being part of its cycles of recovery. Unfortunately, an

[23](Robbins, 1945)

economic system built on the usage of a finite amount of resources could never last forever.

Environmental economists, such as David Pearce and Talbot Page, have been working to expand neoclassical economics to account for pillars of sustainability like intergenerational justice and justice for non-human species. In Page's book, *Conservation and Economic Efficiency: An Approach To Materials Policy* (1977), he employs John Rawls' theory of the "veil of ignorance" to explore an individual's willingness to take action against climate change, by theoretically making them ignorant of what temporal generation they belong to. He argued that placing a rational person under this veil of ignorance forces them to consider the worst-case scenario of belonging to a generation a couple decades into the future, and either for fear of future viability of life on Earth due to climate change or extension of care to the wellbeing of future generations, whoever wore this veil would be more motivated to do their part to ensure that future life will be feasible. David Pearce, in his paper called *Foundations Of An Ecological Economics* (1987), builds on this conceptualization of intergenerational justice to include non-human species as well. He argues that non-human species also have preferences, they prefer certain habitat conditions (like the clams in British Columbia), certain food sources, etc., and says that intergenerational justice would imply sustainability, which implies acknowledging biophysical constraints of the Earth, which would allow non-human species to continue living[24]. Pearce critiques various economic models that propose partial solutions (like taxes on pollution, environmental externalities, etc.) without exhibiting an understanding of intergenerational justice; namely that a truly sustainable

[24](Krishnan et al, 2013)

economy is "ecologically bounded" and must operate *within* biophysical constraints, rather than finding corrective solutions once those thresholds have already been surpassed. With the increasingly dangerous effects of climate change today, it is crucial to look towards alternative kincentric economic frameworks to help find a balance between human activities and the Earth.

3.2 A Kincentric Economy?

Can kincentrism be compatible with economics? Can the two coexist? While there has been an increasing interest among scholars on the topic of Indigenous land and sea management, there is still not adequate Indigenous representation in mainstream climate change discourse and policymaking. So, in order to fill this gap and tackle the economy from an Indigenous point of view, we will return to Dr. Lyla June Johnston's PhD dissertation called *Architects Of Abundance: Indigenous Regenerative Food And Land Management Systems And The Excavation Of Hidden History* (2022). Towards the end of her dissertation, she proposes a theory of Indigenous Regenerative Ecosystem Design (IRED), to unite current research on the topic under a common umbrella, named by an Indigenous person like herself. Her word choices were intentional: *Indigenous* to give credit where it is due, *Regenerative Ecosystem* to emphasize that these strategies improved the functioning and capacities of vast oceans, and forests, not just single plots of land, and finally, *Design,* to make clear that these practices are intentional, Indigenous people really are stewards of the Earth[25]. IRED encompasses all of the ways in which different Indigenous communities have developed sustainable living practices that work in

[25](June Johnston, 2022)

harmony with the Earth and understand the fundamental principles of "reciprocity, reverence, respect, and responsibility to homeland"[26]. Like all models, IRED relies on a few fundamental principles and values:

- Non-humans are equal to or Elder to humans
- Non-human life forms are our relatives
- All lifeforms have an ecological and spiritual role, including humans
- Humans have a sacred covenant to protect and care for their respective homelands
- Humans have a responsibility to create a home for future generations
- Creation is sacred [27]

Dr. Johnston implies that no technological innovation will ever be enough to stop climate change without an unprecedented shift in the world value system altogether, towards a more kincentric way of living. In other words, modern solutions are only partial. A kincentric economy would be an ideal integration of human activities, and nature's boundaries, meaning that instead of damaging the environment, human life would become part of natural lifecycles, eventually bringing Earth's climate to a homeostatic level.

3.3 Indigenous Knowledge Based Solutions

This next section will follow a similar framework to that of the second chapter, it will discuss three related examples of effective Indigenous knowledge-based solutions,

[26]Ibid
[27]Ibid

which have the potential to be scaled up and implemented on a globally systemic level.

3.3.1 Biomimetics – Restoration and Anthropogenic Expansion of Animal Habitat

Remember the clam gardens on British Columbia's Coast? Building intertidal rock walls to augment diverse clam habitats is one of the traditional Indigenous solutions to ensure a secure (and non-invasive) food supply and to reciprocate care back into the Earth by making the living conditions of the local clams more favorable. However, most clam gardens today are eroded remnants, most of which have been washed back into the sea with the rising sea-levels and ocean acidification due to climate change. Restoration of pre-existing clam gardens, bringing sustainable tradition into modernity, is one way to begin to heal humanity's environmental fracture by directly connecting with the sediment, rocks, and water on these coastal areas. Various communities along the Pacific Coast, like the WSÁNEĆ and Hul'q'umi'num Nations on the Gulf Islands, and the Swinomish of Washington, USA, are coming together to restore and monitor ancient clam gardens as they bring them back to life and reconnect with the Earth.

As simple as it sounds, sometimes the best solutions already exist in nature, we just have to pay attention. Biomimicry is defined as "a practice that learns from and mimics the strategies used by living organisms to solve challenges comparable to the ones we face as individuals and societies"[28]. By carefully observing the conditions

[28](Biomimicry Inst., n.d.)

under which a species thrives, humans can fulfill their ecological function and anthropogenically expand animal habitats. While this process has relatively local applications thus far, the same principles of kincentricity can be applied much more generally – to virtually any animal or plant species, in order to ensure their survival, and in turn a renewable food source for humans.

> "Respect is reflected in how clams and other animals are seen as equal to humans. Reciprocity, restraint, and stewardship are reflected in the fact that humans tend to the homes of the clams and ensure sustainable harvest rates. Service and reverence are reflected in the underlying tenet of a divine duty bestowed to humans to care for their non-human counterparts. Efficiency is reflected in the ways clam gardeners let the immense forces of nature do much to the work needed to achieve their goals. Human intervention is valued as an essential piece of the ecological puzzle."[29]

3.3.2 Reforestation Through Direct Seeding in Brazil

In the Brazilian Amazon, a project called the Xingu Seed Network is one of the most relevant examples of climate change mitigation today. To understand the significance of its mission, it is necessary to understand why the Amazon and its Indigenous communities are so central in the fight against climate change. Tropical forests like the Amazon play a crucial role in reversing the damages of climate change through carbon sequestration, maintaining biodiversity, and regulating regional and global climates. The Amazon forest is nicknamed the "lungs of the world" because of its

[29](June Johnston, 2022)

ability to function as an enormous carbon sink; it spans eight countries and has the carbon storage in biomass of 4-14 decades of global emissions[30], making its existence absolutely necessary to decelerate global warming. However, despite its importance, the Amazon is one of the most endangered ecosystems, experiencing losses of thousands of kilometers each year. As this happens, forests lose their resilience, and their ability to regenerate weakens. This cumulative loss of forest coverage poses the threat of forest dieback, the point at which a forest has lost so much biotic life (plants, animals, fungi, bacteria, and other microorganisms), that it will no longer be able to regenerate and the entire forest ecosystem could die. This is the worst case scenario, in which forests like the Amazon convert from carbon sinks to net carbon emitters that further exacerbate global warming.

To prevent this, Brazil has declared certain areas as Permanent Protection Areas (PPAs), including the Xingu River Basin that has been an especially vulnerable target of deforestation due to the expansion of agribusinesses and land privatization in the area. Increased colonial and commercial ventures in the area which, as proven earlier, operate contingent on the depletion of Earth's natural resources, have contributed to further chemical pollution and deforestation around the basin. The Y Ikatu Xingu campaign, translated to "save the good water of the Xingu", began with the main objective of restoring the viability and integrity of the Xingu water and land territory and preventing further deforestation. One of the initiatives of this campaign is the Xingu Seed Network, a primarily woman-run network of around 565 seed collectors, Indigenous and external ecologists and biologists, and research

[30](Durigan, 2013)

institutions like the Instituto Socioambiental (ISA). The network utilizes the Indigenous agroforestry technique of direct seeding whereby a "muvuca" (a mixture) of native plant species are collected and planted throughout deforested areas of the Amazon, guaranteeing that subsequent growth mirrors the strata of native vegetation and thereby embodying kincentric ecology by acting as part of the ecological system. In most cases, manual planting of seeds is still necessary, facilitating the continuous interaction between people, and between people and nature.

The Xingu Seed Network succeeds not only in doing their part to save the Amazon, it builds and fosters a community of Indigenous and non-Indigenous people who are passionate about saving life on Earth and who are willing to educate themselves and dedicate time to do so. There are many obstacles including high costs, legislative pressures and uncertainty, and pressure for Brazil to be competitive in the global economy, but the initiative shows a small-scale example of what a kincentric economy can look like. Due to the growing market for restorative initiatives, the Xingu Seed Network provides financial incomes for seed collectors, showing an alternative income that comes from the well-being of the forest. In a bit under a decade, the network has produced 71 tons of native seeds from 285 regional species, generating US$380 000.00 that was directly transferred to about 300 collectors, from 17 Indigenous communities and 18 rural settlements[31].

[31](Urzedo et al, 2020)

This model of a seeding network has the potential to be implemented on a global scale, in virtually any tropical forest, like those in sub-Saharan Africa, South East Asia and the Pacific, and could be extended to other biomes too. The root of any reforestation project are its seeds, and therefore this is an example of successful application of kincentric ecology for climate and economic justice.

3.3.3 Locally Managed Marine Areas (LMMAs) in the Pacific Ocean

Fiji is home to the Great Sea Reef, which hosts thousands of diverse marine species and habitats, including dolphins, sharks, turtles, all sorts of coral species, and many more that external scientists have not been able to identify yet. Indigenous communities from the area, like the iTaukei, have an embedded principle of kincentric ecology that they call "veisolisoli and kerekere"[32] (give and take) that dictates the relationship that these people have with the ocean and its inhabitants. Fiji has many commonly used fishing grounds, but they can also attract illegal fishers who do not abide by the same standards of Indigenous marine stewardship and threaten the health of its vast ecosystem.

> "These cultural beliefs [kincentric ecology] affect resource allocations and the potential for responsible environmental stewardship of these property rights regimes contrasts markedly with the pitfalls of the western open access approaches."[33]

[32] (WWF, 2006)
[33] Status and potential of locally-managed marine areas in the Pacific Island Region: meeting nature conservation and sustainable livelihood targets through wide-spread implementation of LMMAs

Overfishing, poisoning, and poaching has threatened the survival of various species in the reef and surrounding areas and has led to a network of Fiji Locally Managed Marine Areas (FLMMAs) which is the collaborative work of local village chiefs in Fiji and their communities. These areas are called "waitui tabu", or no-take zones where harmful fishing techniques (dynamite fishing and cyanide fishing), poaching from Marine Protected Areas, and dumping waste are banned, creating conditions under which aquatic life can recover and regenerate. Fiji's initial pledge was to cover 30% of its territorial waters by 2020[34]. While they have yet to achieve this, Fiji is working with government agencies, global partners like the International Union for Conservation of Nature (IUCN), and the local Indigenous communities to make this happen as soon as possible.

The Fijian example of restricting free access to surrounding waters is a move towards better protection of the health of its ecosystems. This general structure is one that can be applied to any body of water, alongside other complementary regulation, such as consistent monitoring or habitat restoration projects. Furthermore, the distinction of FLMMAs allows Indigenous communities to continue their role as stewards of the land and sea and is an example of kincentric values shown through environmental regulation. As climate change continues to threaten island countries like Fiji, it is crucial that governmental regulation starts to reflect values of restraint when it comes to consumption of natural resources – on a global level.

[34] (WWF, 2006)

CONCLUSION

What influences humanity's relationship to the Earth? How does this impact the way that we perceive climate change? The primary argument of this research paper is that Indigeneity is the most relevant determinant of the nature of a human's relationship to the Earth, and that some of the threats to the continuation of this relationship are colonization and economic gain. By putting into discussion ideas of Indigeneity, colonization, and the Earth, this thesis aims to explain the deeper causes of anthropogenic climate change.

Beginning with a discourse on the various definitions and understandings of "Indigeneity", the goal was to flesh out something that is more often than not, taken for granted. We all come from somewhere, but less often do we examine how this influences our ties to the natural world, and thus our perception of climate change. If it is human activities that have spun the world's climate out of balance, then what influences our human activities, and what enables them? Back to Robbins' definition of economics, it is the nature of those "*ends*" that humans pursue that dictates the relationship between humans and the Earth's resources.

Throughout this paper, kincentric ecology represents the conceptual understanding of love in the context of climate change. The kinship that it describes between people and each other, and people and all other forms of life is one that characterizes most Indigenous communities and shapes a way of life that is compatible and harmonious with the Earth. As such, Indigenous communities contribute to an important group of solutions to climate change – kincentric

solutions. Therefore, it is extremely important for Indigenous knowledge and history to be taken into account when drafting global solutions against climate change, as failing to do so will also fall short of addressing its fundamental causes. However, there is a barrier between Indigenous solutions and systemic implementation. One must acknowledge that the makeup of the land in present-day is much different than what Indigenous communities managed locally. In addition to this, there are century-old stigmas surrounding Indigenous communities that prevent their representation in mainstream conversations and committees about climate change, making it harder to apply kincentric values to global solutions.

With the reflections and experiences shared throughout this paper, a new alternative of life on Earth is presented, one that can promise a healthy Earth for all the generations to come.

> "There is a word in my Diné language: *Hozhó*. *Hozhó* is the joy of being a part of the beauty of all creation. When we understand that humans are an expression of the earth's beauty, we understand that we, too, have an ecological role. *Hozhó* understands that we belong here. *Hozhó* understands that the earth needs us. When we become her friend, her ally, her partner in life— instead of her dominator, her "superior," or her profiteer—we can transform dead systems to living ones. This does not involve isolating national parks and never touching a blade of grass. It involves rolling up our sleeves, living within her processes, becoming a part of the Earth sys- tem, and using our evolved brains to serve life, rather than profit from life, on holistic,

regional scales. If our Ancestors around the world proved this is possible, there is strong hope that we can do it again."[35]

[35](June, 2022)

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